

Digital Systems

Tutorial 10

May 2023

1. **Word number 723 in a 1024×16 memory contains the binary equivalent of 3,451. List the 10-bit address and the 16-bit memory content of the word. Also write the address and memory content in hexadecimal form.**

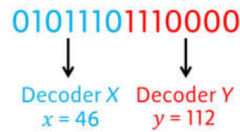
SOL.

	Decimal	Binary	Hex.
Address	723	1011010011	2D3
Memory content	3,451	0000110101111011	D7B

2. **A $16K \times 4$ memory uses coincident decoding by splitting the internal decoder into X-selection and Y-selection.**
 - (a) Name one advantage of using coincident decoding in lieu of memory decoding with a single decoder.
 - (b) What is the size of each decoder, and how many AND gates are required for decoding the address?
 - (c) Determine the X and Y selection lines that are enabled when the input address is the binary equivalent of 6,000.

SOL.

- (a) Coincident decoding allows for architectures with less AND gates and, consequently, more economical designs.
- (b) 16K bits amounts to $16 \times 2^{10} = 2^{14}$ bits; distributing data processing evenly, each decoder will carry $2^7 = 128$ bits. Each decoder will have 7 inputs. Each decoder has 128 AND gates, totaling 256 AND gates for two decoders.
- (c) Converting 6,000 to binary form, we obtain 101101110000. Adding one extra zero to turn this into a 14-bit string, we write 0101101110000. This input is to be partitioned into two words, each of which is to be assigned to one of the two decoders. The address lines enabled upon entering this data are found to be 46 and 112, as shown.



3. Consider a memory system that takes 25 nsec to service the access of a single 64-bit word. what is the bandwidth of the processor-memory interface(in Mbytes per second) ?

SOL. 8 bytes of data can be accessed in 25 nsec. So in 1 second, number of bytes accessed = $8 / (25 * 10^{-9}) \text{ bytes} = 320 \text{ Mbytes}$

4. Ramnarayan designed a 1M (2^{20}) bit memory. During testing, one of the bit out of 1M bits got damaged. Ramnarayan now has to find the location of the damaged bit. To do this he has to perform the following test.

The memory is plugged into an IC tester, which access each bit one-by-one sequentially starting from the bit at location 00000H. Now, at each bit, the tester first writes/stores high (3V) to the memory and then, reads the memory.

The time taken to access the memory/bit: 1 micro second.

The time taken to write to the memory: 1 micro second.

The time taken to read the memory: 1 micro second.

The bit which don't read correctly is designated faulty and tester blows an alarm. If the alarm blew exactly after 1 sec, find the location of the damaged bit.

The complete operation of accessing, writing and read a memory location is completed in $(1\mu + 1\mu + 1\mu) = 3\mu\text{sec}$

The time taken to read 2^{20} memory locations

$$= 2^{20} \times 3\mu\text{sec}$$

$$= 3.145728 \text{ sec}$$

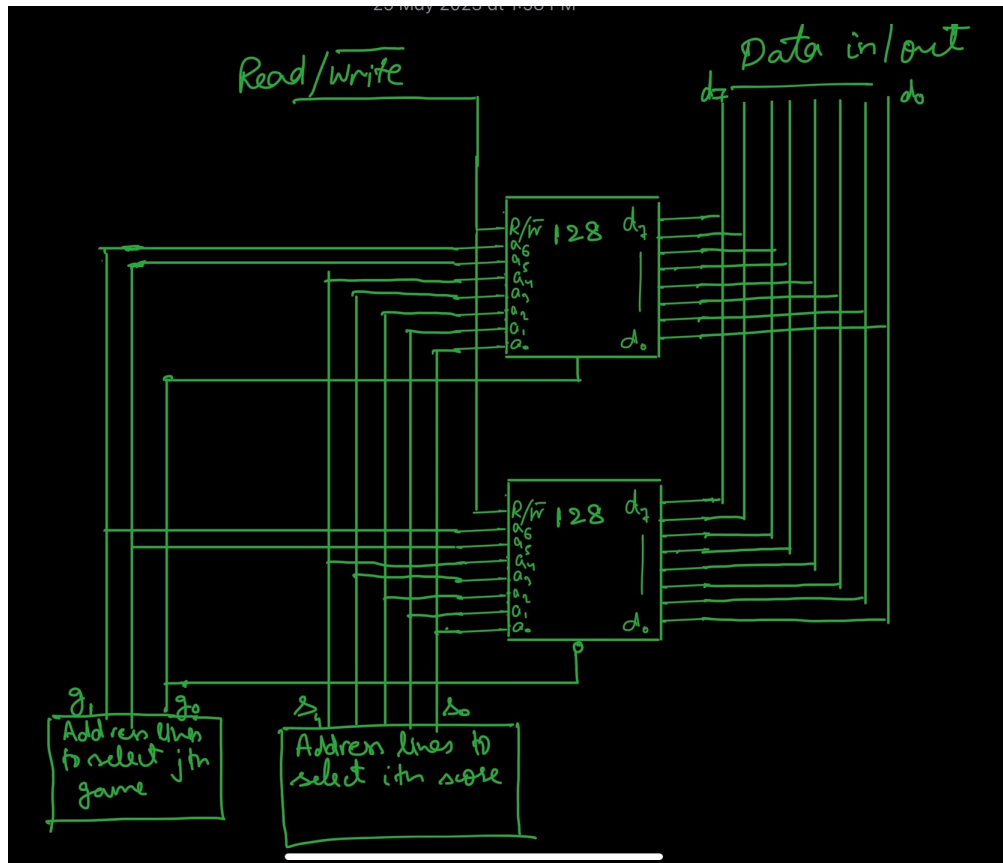
The alarm blew up at 1sec

$$\Rightarrow \text{The memory location} = \frac{2^{20} \times 3\mu\text{sec}}{3.145728}$$

$$= 333333.33$$

$$= 333334 \text{ th location}$$

5. Ramnarayan likes to play 5 different games. He wants to remember/store the score of the last 25 matches played in each game. He don't have a notebook or pen, but He is an electronics engineer. So, he decides to store the desired scores in the memory ICs available to him. He has two nos 128 byte memory ICs available. Design the Memory circuit to enable the read and write data to and from the memories. What are the disadvantages of this circuit?



6. The excitation function describes an asynchronous sequential circuit

$$Y = x_1x'_2 + (x_1 + x'_2)y,$$

and the output is $z = y$

- Draw the logic diagram of the circuit.
- Derive the transition table and output map.
- Obtain a two-state flow table.

Sol. (a) Logic diagram of the circuit,

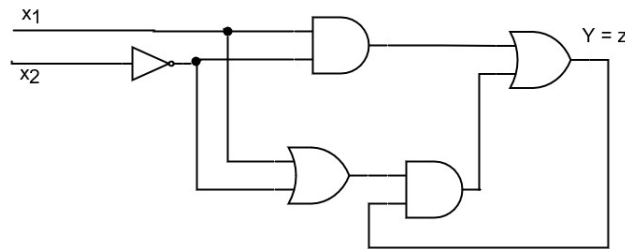


Figure 1

- Transition table and output map,
- and two-state flow table,

$x_1 \ x_2$	00	01	11	10
0	0	0	0	1
1	1	0	1	1

(a) Map for Y

$x_1 \ x_2$	00	01	11	10
0	0	0	0	1
1	1	0	1	1

(b) Transition Table

$x_1 \ x_2$	00	01	11	10
a	a	a	a	b
b	b	a	b	b

(c) Flow Table

Figure 2

7. Derive the transition table for the asynchronous sequential circuit shown in figure 1. Determine the sequence of internal states Y_1 and Y_2 for the following input sequence x_1x_2 : 00, 10, 11, 01, 11, 10, 00.

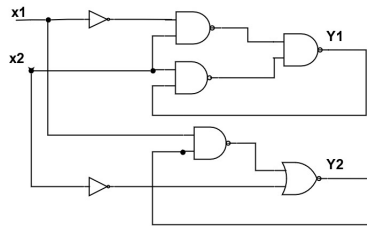


Figure 3

Ans. (a) Transition table for given circuit,

$x_1 x_2$	00	01	11	10
$y_1 y_2$	00	10	00	00
00	00	10	01	00
01	00	10	11	00
11	00	10	10	00
10	00	10	10	00

(a) Transition Table

Figure 4

(b) Sequence of internal states of Y_1 and Y_2

$x_1 x_2$	$y_1 y_2$
00	00
10	00
11	00
01	10
11	10
10	00
00	00

Figure 5

8. Consider the reduced flow table shown in the following figure.

- Obtain the transition table and output map.
- Derive the digital logic circuit.

		Inputs			
		00	01	11	10
states	a	a, 0	a, 0	b, 0	a, 0
	b	d, 0	a, 0	b, 0	c, 1
	c	d, 0	a, 0	b, 0	c, 1
	d	a, 0	a, 0	a, 0	a, 0

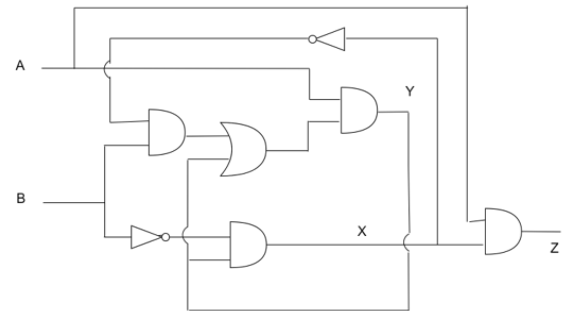
Transition Table

		Inputs AB			
		00	01	11	10
states xy	00	00	00	01	00
	01	10	00	01	11
	11	10	00	01	11
	10	00	00	00	00

Output Map

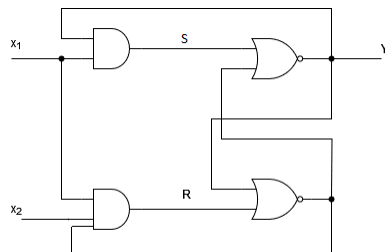
		Inputs AB			
		00	01	11	10
states xy	00	0	0	0	0
	01	0	0	0	1
	11	0	0	0	1
	10	0	0	0	0

$Z = yAB'$



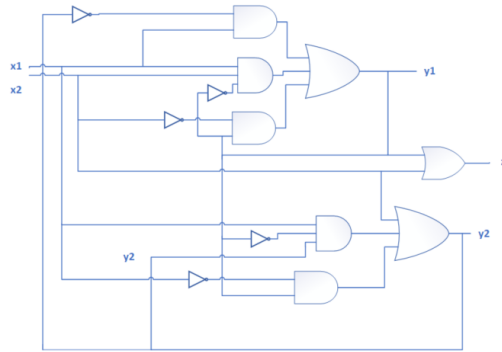
Sol.

9. Comment on the stability of the circuit represented by the following state diagram.



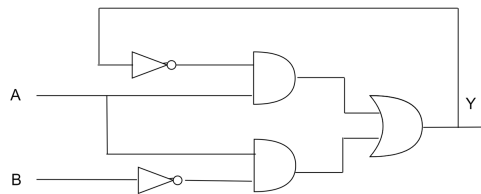
Sol: Circuit is unstable.

10. Comment on the stability of the following circuit.



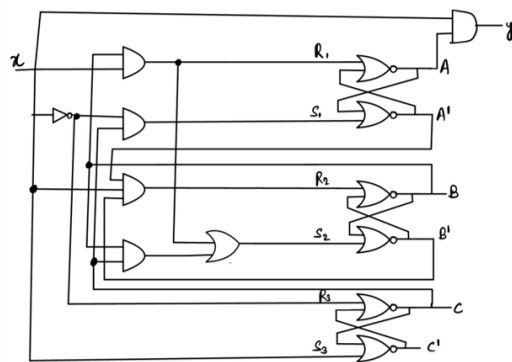
Sol: Circuit is unstable.

11. In the circuit given assume both the inputs A and B are equal to 1. Each gate has a propagation delay of 15 nanoseconds. Find the frequency of oscillation of the circuit.



Sol : Frequency of oscillation= 11 MHz

12. Derive the transition table and output map for the following asynchronous sequential circuit.



Sol :

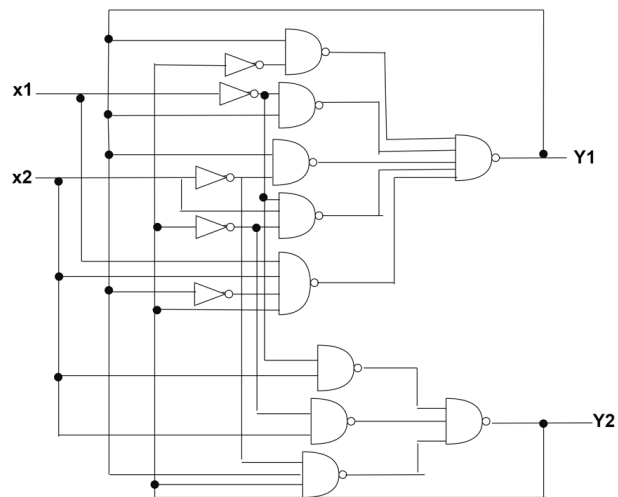
Transition Table

		Inputs x	
		0	1
states abc	000	000	001
	001	100	001
	011	110	011
	010	010	011
	100	100	101
	101	100	101
	111	110	011
	110	110	011

Output Map

		Inputs x	
		0	1
states abc	000	0	0
	001	0	0
	011	0	0
	010	0	0
	100	0	1
	101	0	1
	111	0	1
	110	0	1

13. Find the function performed by the following circuit when input x_2 changes from 0 to 1. Assume input x_1 is at logic 1. (Hint: Use transition table)



Sol : Circuit acts as a MOD-3 up counter.

1 Practice Questions

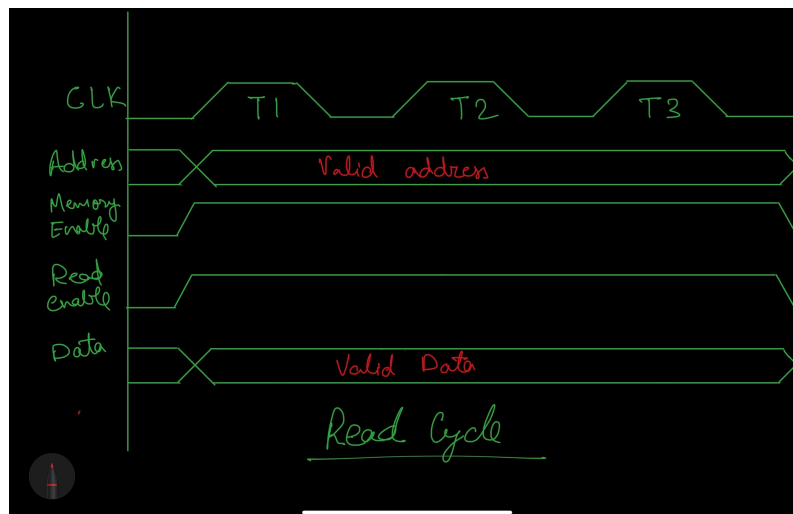
1. A RAM chip has a capacity of 1024 words of 8 bits each. The number of 2-to-4 decoders with enable lines needed to construct a 16K x 16 RAM system will be? and also represent with block diagrams

SOL.

To construct a memory system with 16K words using 1024x8 RAM chips, we have to connect 16 such chips in parallel. The decoder will have to select one of the 16 chips. To construct a 4x16 decoder, we require 5, 2x4 decoders.

2. Draw the read cycle for accessing the memory requiring 3 clock cycles with active high memory and read enable.

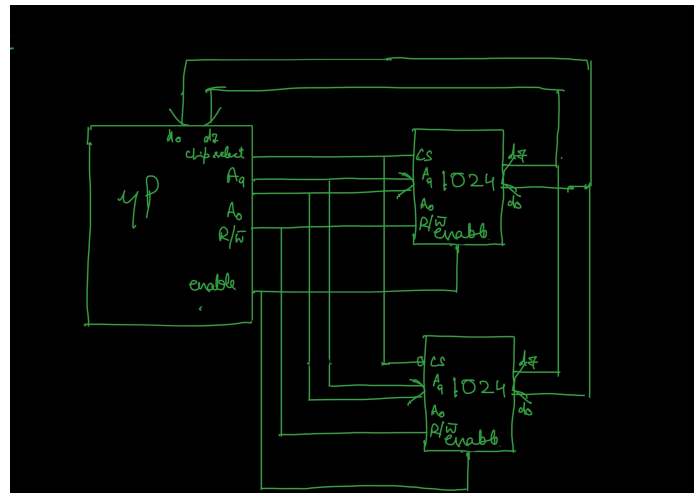
SOL.



3. A microprocessor has 8-bit data bus, 10-bit address bus and 2 chip-select/enable pins. Determine the maximum number of memory locations (in bits and bytes) this μ Processor can access.

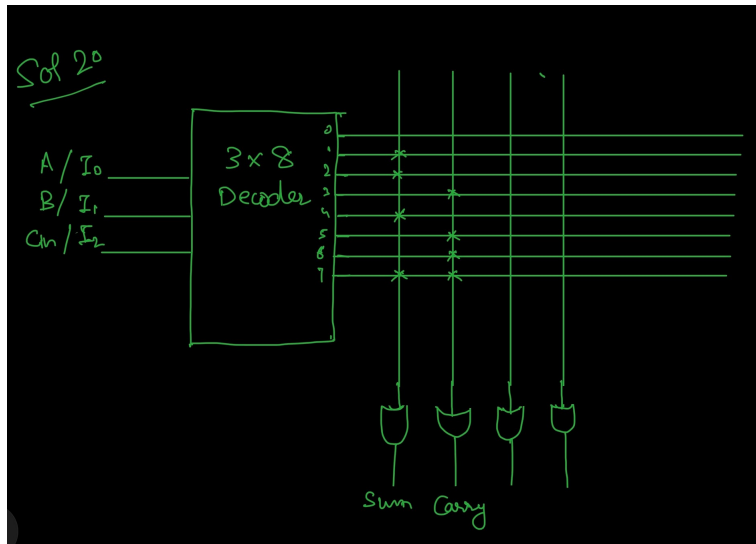
8 bit data bus
 10 bit address bus
 2 chip select / enable pins
 $\Rightarrow$ using 10 pins of address bus and 2 pins of chip select / enable pins, we can access $2^{12} = 4096$ memory locations. / bytes
 $\Rightarrow 4096 \times 8 = 32768$ bits

4. Design the circuit diagram for accessing the 2048 bytes for the above μ Processor. Using 2 nos 1024 byte memory.



5. Design a Full-adder and subtractor using ROM.

above design is for full adder similarly can be design for subtractor.



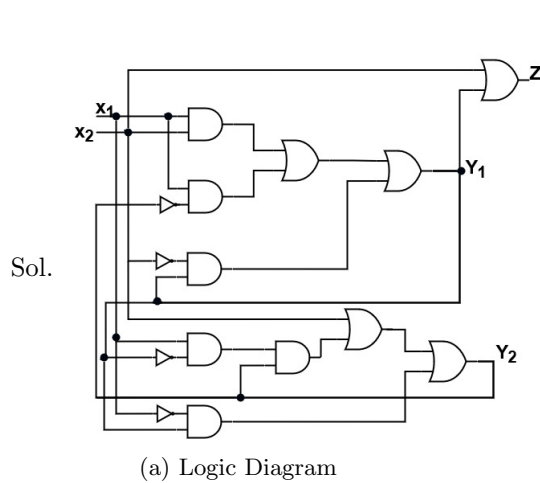
6. An asynchronous sequential circuit has two internal states and one output. The two excitation functions and one output function describing the circuit are, respectively,

$$Y_1 = x_1x_2 + x_1y_2' + x_2'y_1$$

$$Y_2 = x_2 + x_1y_1'y_2 + x_1'y_1$$

$$z = x_2 + y_1$$

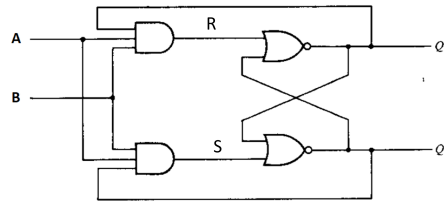
- (a) Draw the logic diagram of the circuit.
 (b) Derive the transition table and output map.



		$x_1 x_2$			
		00	01	11	10
$y_1 y_2$	00	00	01	11	10
	01	00	01	11	01
	11	11	01	11	10
	10	11	01	11	10

(b) Transition Table

7. Obtain the transition table for the following circuit and comment on the stability of the circuit. Assume both A and B are equal to 1.



Sol : Circuit is unstable.